Homological Algebra Week1 Solutions

Exercise 1

For any map φ, ψ where compositions $f \circ \varphi, f \circ \psi$ make sense, we have if f is injective then $f \circ \phi = f \circ \psi$ implies $\phi = \psi$ as maps between sets. Thus the morphism $f\circ$ is injective. Similarly, as g is surjective $\circ g$ is an injective morphism. For the exactness as $g \circ f = 0$ the inclusion,

$$\operatorname{im} f\circ \subseteq \ker g\circ$$

is clear. If $\varphi : X \to B \in \ker g\circ$ then $\operatorname{im} \varphi \subseteq \ker g = \operatorname{im} f$. Thus we can assume $B = \operatorname{im} f$, namely f is bijective. Set $\varphi' = f^{-1} \circ \varphi$. This is in $\mathrm{Hom}_{\mathrm{Ab}}(X, A)$ and $f \circ \varphi' = f \circ f^{-1} \circ \varphi = \varphi$.

For the other case, by similar argument, we have,

$$\operatorname{im} \circ g \subseteq \ker \circ f.$$

Let $\psi \in \ker \circ f$. Similarly, $\ker \psi \supseteq \operatorname{im} f = \ker g$. Thus we consider $B/\ker g$ instead of g. This make g bijective, therefore set $\psi' = \psi \circ g^{-1} \in \mathrm{Hom}_{\mathrm{Ab}}(C, X)$.